



**POLYTECHNIQUE
MONTRÉAL**

UNIVERSITÉ
D'INGÉNIERIE

LOG8430E : TP2

Comparison of NoSQL Databases

Introduction

- **Context** : Compare the performance of three distributed NoSQL databases using the YCSB benchmark
- **Databases studied** :
 - MongoDB (document-oriented)
 - Redis (key-value)
 - Apache Cassandra (column-oriented)
- **Format**: Report must use the provided LaTeX template (PDF submission, IEEE format, max 5 pages)
- **Deadline**: Saturday, December 6th, before 23h59
- **Important** : Experiments must be completed before the presentation to report actual results.

To get started

- Install Docker Desktop (<https://www.docker.com/products/docker-desktop/>)
- Install the YCSB release
- Write a .yml file for each database (e.g., 3-node Redis cluster).
- Place the Redis .yml file in the YCSB bin folder for convenience.
- Start the cluster using `docker-compose`

How to get started

Command to start Redis Cluster:

`docker compose up -d`

- Starts all containers defined in your .yml file in detached mode (-d)
- Each Redis node (master and replicas) will start according to the configuration

Commands to verify cluster communication

`docker exec -it redis-master sh`
`redis-cli`
`set test "hello"`

- `docker exec -it redis-master sh`: opens a shell inside the Redis master container.
- `redis-cli`: launches the Redis command-line interface.
- `set test "hello"`: writes a key-value pair to the master.

How to get started

Command to verify cluster communication :

```
Docker exec -it redis-replica-1 sh  
redis-cli -p 6380  
get test
```

- Connects to the first replica (redis-replica-1) on port 6380
- get test : retrieves the value from the replica to verify replication (hello).

Load Data with YCSB

Command to load data into Redis:

```
.\ycsb load redis -s -P ..\workloads\workloada -p "redis.host=127.0.0.1" -p  
"redis.port=6379"
```

- Loads data into Redis according to *workloada* workload.
- -s: prints status during execution.
- -P: points to the workload file.
- -p "redis.host=127.0.0.1" and -p "redis.port=6379": connect to the Redis instance.

Run YCSB Workload

Commands to run benchmark on Redis

.\ycsb run redis -s -P ../workloads/workloada -p "redis.host=127.0.0.1" -p
"redis.port=6379"

- Runs the workload defined in workloada on Redis.
- Measures throughput and latency for read/write operations.
- Repeat 10 times for statistical confidence.

Flush Database

Commands to Commands to run benchmark on Redis

```
docker exec -it redis-master redis-cli  
> FLUSHALL  
> exit
```

- FLUSHALL: clears all keys in all databases on the master.
- Ensures no leftover data from previous workloads affects new tests.